Module 1 : Introduction to NLP

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Agenda

- Course Objectives
- Course Outcomes
- Prerequisites
- Text / Reference Books
- Assessment Methods
- Introduction to NLP

Course Objectives

- To define natural language processing and to learn **various stages of natural language processing**.
- To describe basic concepts and algorithmic description of the main language levels: **Morphology, Syntax, Semantics, and Pragmatics & Discourse analysis**.
- Learn and apply both classical statistical and modern neural approaches to NLP, including **sequence models and transformer architectures.**
- **Design, implement, and evaluate NLP and generative AI systems** for real-world applications
- Analyze and address **ethical, social, and technical challenges in NLP and generative AI**, including bias, fairness, privacy
- To learn advanced NLP techniques for developing real **world NLP applications using LLM** to solve real-world language processing problems

Course Outcomes

- Have a broad **understanding of the field of natural language processing**.
- Apply NLP techniques like **preprocessing, POS tagging, parsing, and semantic analysis.**
- Implement and fine-tune **statistical and neural models (RNNs, LSTMs, transformers)** for NLP tasks.
- Design, implement and evaluate **LLMs for text generation, summarization, translation, and chatbots.**
- Design prompts and apply **prompt engineering to optimize LLM outputs**.
- Design prompts and apply prompt engineering to optimize LLM outputs to solve real-world language processing challenges.

Prerequisites

- Python Programming,
- Data structures & Algorithms,
- Machine Learning Concepts,
- Probability and Statistics,
- Theory of Computation

Text Books

1. Daniel Jurafsky, James H. Martin "Speech and Language Processing" Second Edition, Prentice Hall, 2008.
2. Julien Chaumond, Hamza Tahir, Antania Guli," LLM Engineer's Handbook", Packt Publication
3. Natural Language Processing with Transformers: Revised Edition by Lewis Tunstall, Leandro von Werra, and Thomas Wolf
4. Prompt Engineering and ChatGPT , Russel Grant (Author), Jeremy Diener
5. Karthikeyan Sabesan , Sivagami sundari , Nilip Dutta ,"Generative AI for Everyone: Deep learning, NLP, and LLMs for creative and practical applications"
6. Christopher D.Manning and Hinrich Schutze, " Foundations of Statistical Natural Language Processing ", MIT Press, 1999.

Reference Books

1. Siddiqui and Tiwary U.S., Natural Language Processing and Information Retrieval, Oxford University Press (2008).
2. Daniel M Bikel and Imed Zitouni " Multilingual natural language processing applications" Pearson, 2013
3. Alexander Clark (Editor), Chris Fox (Editor), Shalom Lappin (Editor) " The Handbook of Computational Linguistics and Natural Language Processing " ISBN: 978-1-118-78
4. Nitin Indurkhya and Fred J. Damerau, — Handbook of Natural Language Processing, 2nd edition, Chapman and Hall / CRC Press, 2010
5. Anjanava Biswas , Wrick Talukdar ,'Building Agentic AI Systems: Create intelligent, autonomous AI agents that can reason, plan, and adapt.
6. Steven Bird, Ewan Klein and Edward Loper, Natural language processing with Python: analyzing text with the natural language toolkit, O‗Reilly Media, 2009.

Assessment Methods

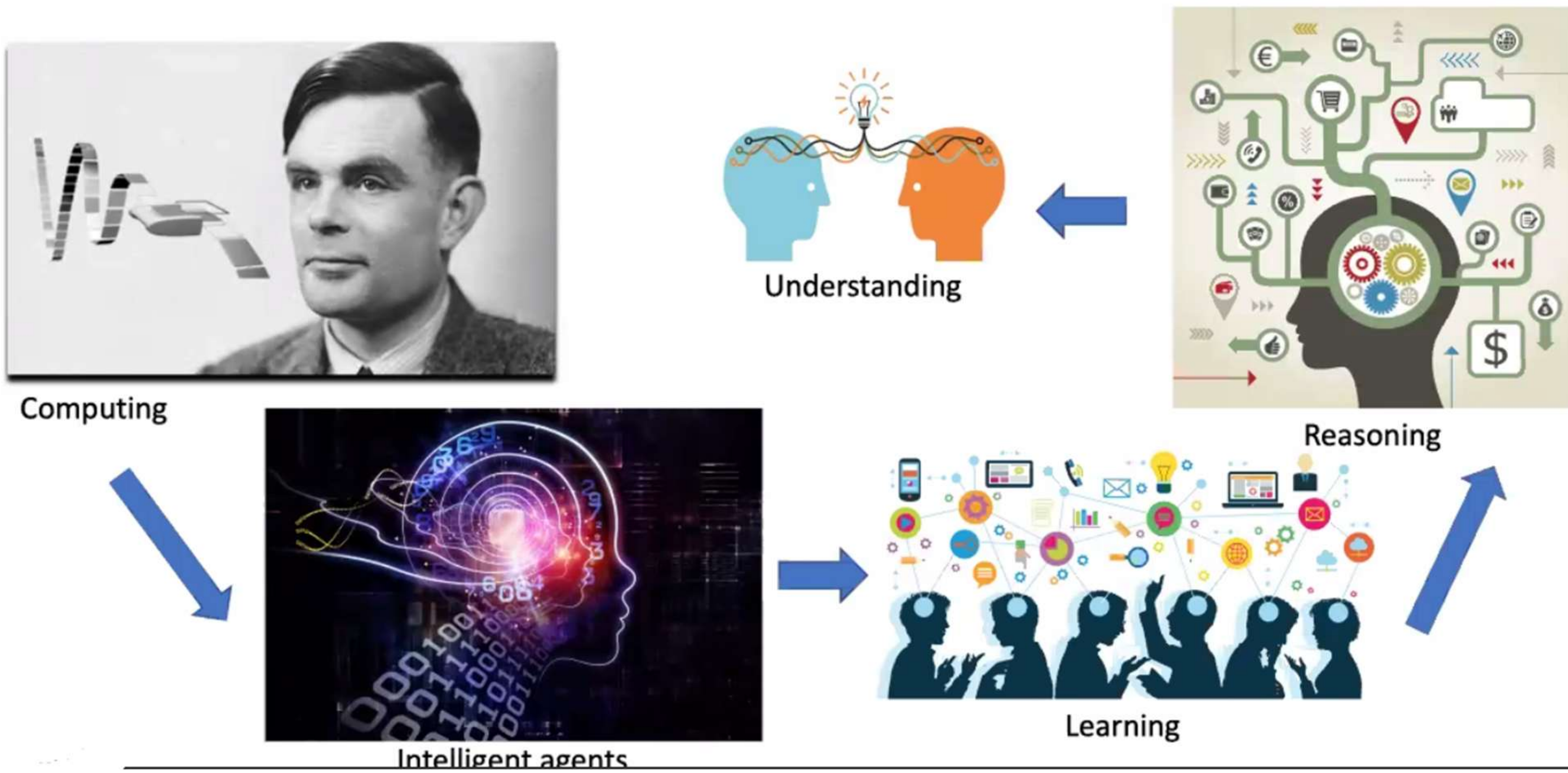
- Internal Assessments :
  - Mid-Term Test : 20 marks
  - Continuous Assessment : 20 marks
- End Semester Examination : 60 marks

Module 1 : Introduction to NLP

- Introduction to NLP
- Origin & History of NLP
- Language, Knowledge and Grammar in language processing
- Stages in NLP
- Ambiguities and its types in English and Indian Regional Languages
- Challenges of NLP
- Applications of NLP

Source: Daniel Jurafsky, James H. Martin “Speech and Language Processing” Third Edition, Prentice Hall,

What is Artificial Intelligence?



Application Areas of Artificial Intelligence

NATURAL LANGUAGE

SPEECH

VISION

ONTOLOGICAL REASONING

COGNITIVE REASONING

... AND SO ON

As a running example of an AI Application Area

Natural Language: The Final Frontier

- Natural Language Processing
- Natural Language Understanding
- Natural Language Generation

Natural Language Processing

- RECOGNIZING PARTS OF SPEECH
- PARSING A SENTENCE
- RECOGNIZING NAMED ENTITIES
- RETRIEVING TEXTUAL INFORMATION
- CLASSIFYING AND CLUSTERING TEXTUAL INFORMATION

Natural Language Processing (NLP)

Natural Language Understanding

- Taking some spoken/typed sentence and working out what it means

Natural Language Generation

- Taking some formal representation of what you want to say and working out a way to express it in a natural (human) language (e.g., English)

Motivation for NLP

- Understand language analysis & generation
- Communication
- Language is a window to the mind
- Data is in linguistic form
- Data can be in Structured (table form), Semi structured (XML form), Unstructured (sentence / multi-media form).

Goals : NLP

- Have some understanding based on communication with Natural Language and not Java, C++, Perl, ...
- In order to receive and send information in ways easily understandable by human users
- Ultimate goal: Natural human-to-computer communication
- Sub-field of Artificial Intelligence, but very interdisciplinary

How to Get There....

- NLP applications are all similar in that they require some level of understanding.

- Understand the query, understand the document, understand the data being communicated...

What is NLP?

- **Wiki: Natural language processing (NLP)** is a field of computer science, artificial intelligence, and computational linguistics concerned with the interactions between computers and human **(natural)** languages.

Go Beyond the Keyword Matching...

- Identify the structure and meaning of words, sentences, texts and conversations
- Deep understanding of broad language
- NLP is all around us

Natural Language Understanding

Users interact in a way that most preserves their natural language

- Culture
- Geography
- Economy
- Accent
- Dialect
- Belief systems

Social and temporal norms

- "Heyo!"

Natural Language Generation

The holy grail

NLP + NLU

History : NLP contd..

- **1950s**
  - Early MT: word translation + re-ordering
  - Chomsky's Generative grammar
  - Bar-Hill's argument

- **1960-80s**
  - Applications
    - BASEBALL: use NL interface to search in a database on baseball games
    - LUNAR: NL interface to search in Lunar
    - ELIZA: simulation of conversation with a psychoanalyst
    - SHREDLU: use NL to manipulate block world

History : NLP

- Methods
  - ATN (Augmented Transition Networks): extended context-free grammar
  - Case grammar (agent, object, etc.)
  - DCG – Definite Clause Grammar
  - Dependency grammar: an element depends on another

- **1990s-now**
  - Statistical methods
  - Speech recognition

NLU : Language, Knowledge & Grammar

Raw speech signal

↓

- **Speech recognition**

Sequence of words spoken

↓

- **Syntactic analysis** using knowledge of the grammar

Structure of the sentence

↓

- **Semantic analysis** using info. about meaning of words

Partial representation of meaning of sentence

↓

- **Pragmatic analysis** using info. about context

Final representation of meaning of sentence

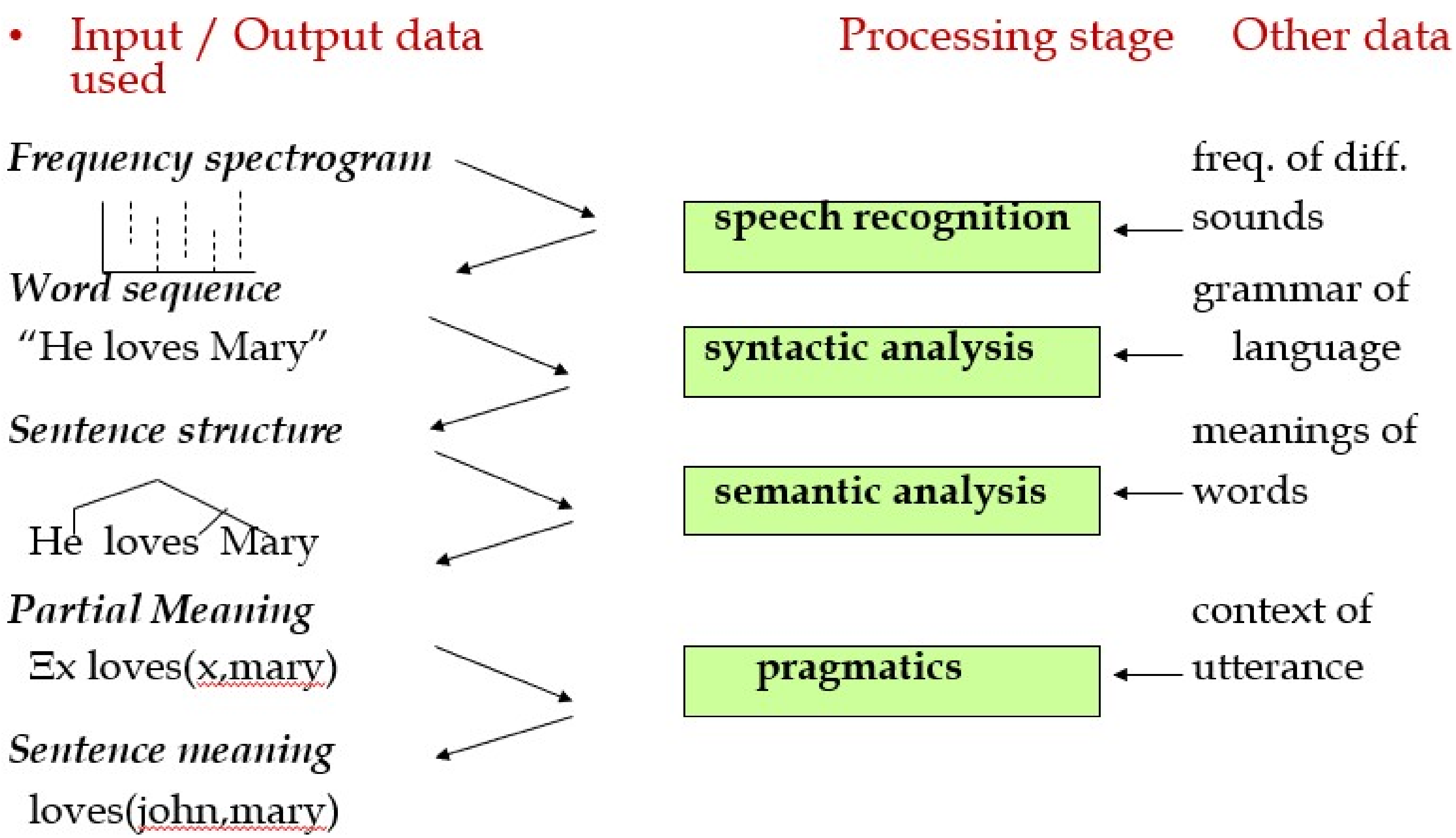
NLU : Language, Knowledge & Grammar

Language levels:

- Phonology
- Morphology
- Syntax
- Semantics
- Pragmatics

Knowledge representation enables meaning understanding

NLU : Language, Knowledge & Grammar



Grammar in Language Processing

Formal grammars:

- Regular Grammar
- Context-Free Grammar (CFG)
- Dependency Grammar

Used for parsing and structure analysis

Grammar in Language Processing

Formal grammars: Regular Grammar

•Regular Grammar is a formal grammar that generates regular languages, used to model simple, rule-based language structures.

Why It Matters in NLP:

- Enables pattern recognition in text (tokens, phrases, entities)
- Forms the backbone of Finite State Machines (FSMs) and Regular Expressions (REs)
- Supports Lexical analysis, morphological processing, and text validation.

Key Characteristics:

- Rules are in the form of A-> aB or A->a
- Efficient for fast parsing and matching
- Ideal for handling structured and semi structured text.

Real-World Applications:

- Chatbots and Speech Systems: Recognize commands and intent
- Information Retrieval: Index and retrieve keyword patterns
- Data Cleaning: Validate formats like emails, IDs, and dates
- Tokenization: Identify word boundaries and linguistic units.

Grammar in Language Processing

Formal grammars: Context Free Grammar (CFG)

•A Context-Free Grammar defines **hierarchical syntactic structures** of language using production rules of the form **A → α**, where α can be a mix of terminals and non-terminals.

Why It Matters in NLP:

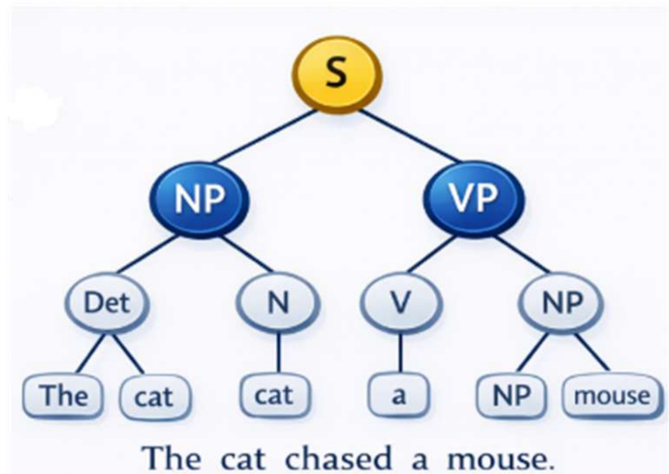
- Drives **syntactic parsing** and **sentence structure analysis**
- Essential for **Natural Language Understanding (NLU)** and **machine translation**
- Powers **grammar-based language modeling** and **semantic interpretation**

Key Characteristics:

- Captures **nested and recursive language patterns**
- More expressive than Regular Grammar
- Foundation for **parse trees** and **derivation analysis**

Real-World Applications:

- **Chatbots & Virtual Assistants:** Understand complex user queries
- **Information Extraction:** Identify subject–verb–object relationships
- **Machine Translation:** Map source to target grammar structures
- **Speech & Language Systems:** Improve syntactic correctness and fluency.



Grammar in Language Processing

Formal grammars: **Dependency Grammar**

•Dependency Grammar represents sentences as a set of **binary relations (dependencies)** between words, where each word depends on a **head word**, forming a tree-like syntactic structure.

Why It Matters in NLP:

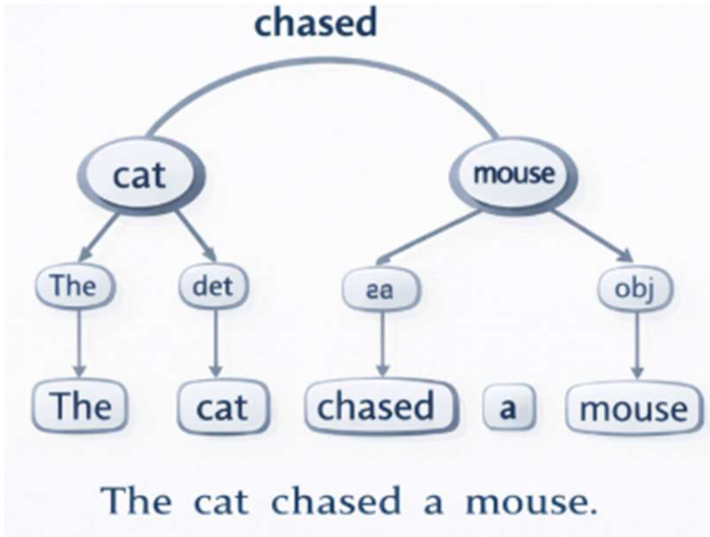
- Enables **direct word-to-word syntactic mapping**
- Critical for **semantic role labeling** and **relation extraction**
- Improves **machine translation, question answering, and text analytics**

Key Characteristics:

- Focuses on **grammatical relationships instead of phrase structure**
- Produces **dependency trees** showing head–dependent links
- Efficient for **handling free word-order languages**

Real-World Applications:

- **Chatbots & Assistants:** Understand intent through grammatical relations
- **Information Extraction:** Identify subject–object–verb dependencies
- **Search Engines:** Enhance query understanding and ranking
- **Language Translation:** Align syntactic roles across languages



Forms of Natural Language

- The input/output of a NLP system can be:
  - **written text (Mostly concerned with this)**
  - **Speech (More challenges in Speech Recognition)**

- To process written text, we need:
  - **lexical, syntactic, semantic knowledge about the language**
  - **discourse information, real world knowledge**

NLP : Terminology

Phonetics and phonology	The study of language sounds
Ecology	The study of language conventions for punctuation, text mark-up and encoding
Morphology	The study of meaningful components of words
Syntax	The study of structural relationships among words
Lexical semantics	The study of word meaning
Compositional semantics	The study of the meaning of sentences
Pragmatics	The study of the use of language to accomplish goals
Discourse conventions	The study of conventions of dialogue

Levels of Natural Language Processing

- Level 1 – Speech sound (Phonetics & Phonology)
- Level 2 – Words & their forms (Morphology, Lexicon)
- Level 3 – Structure of sentences (Syntax, Parsing)
- Level 4 – Meaning of sentences (Semantics)
- Level 5 – Meaning in context & for a purpose (Pragmatics)
- Level 6 – Connected sentence processing in a larger body of text (Discourse)

Levels of NLP : Example

L1 : sound

L2 : Dog - Dog(*s*), Dog(*ged*), Lady – Lad(*ies*)

Should we store all forms of words in the lexicon?

L3 : Ram goes to market ***(right)***

goes Ram to the market ***(wrong)***

L4 : translation from unstructured to structured representation

go : (event); *agent* : Ram; *source* : ? ; *destination* : market

Levels of NLP : Example

- L5 : User situation & context

  "Is that water?" – the action to be performed is different in a chemistry lab and on a dining table.

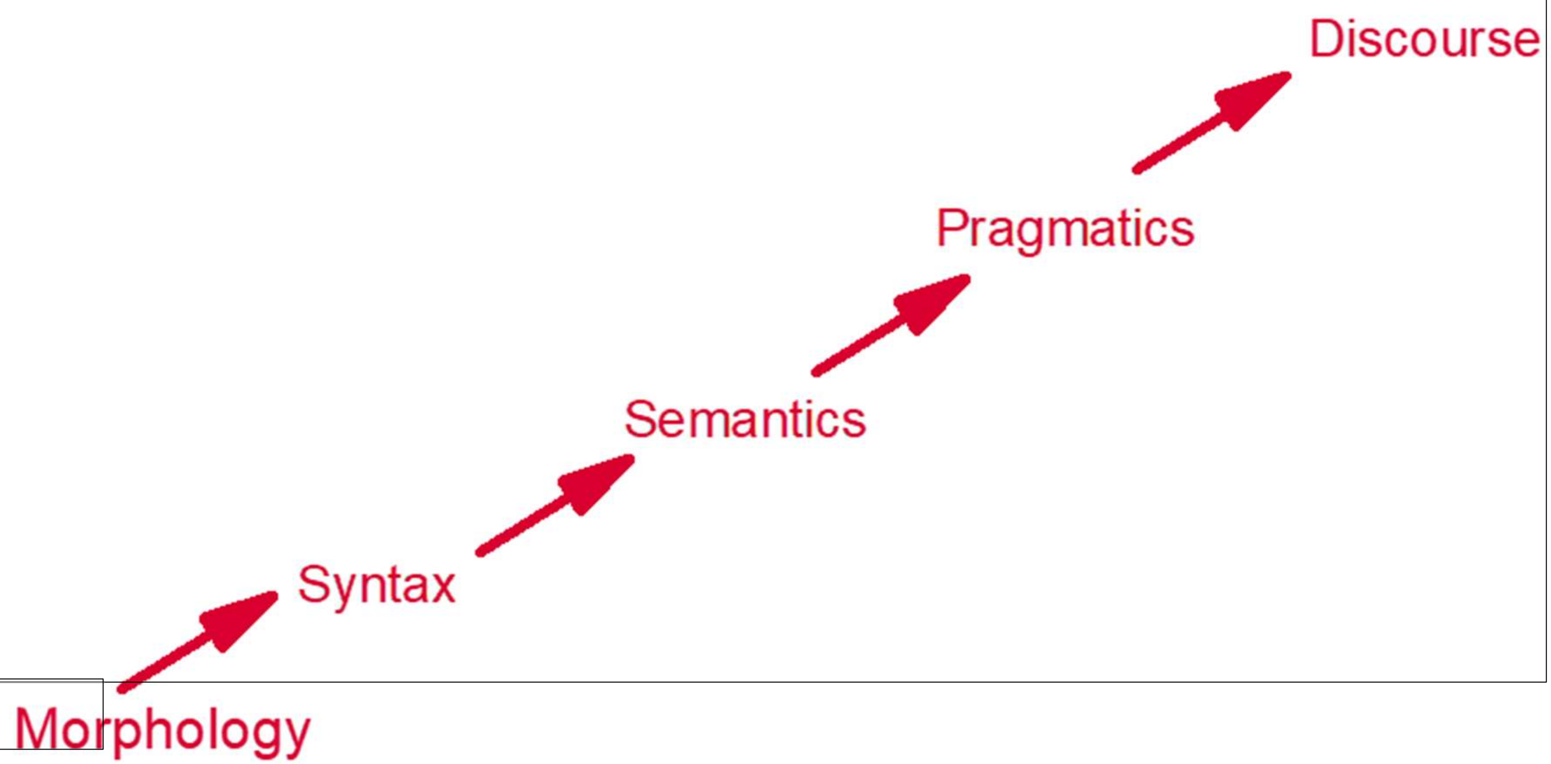
- L6 : Backward & forward references –
- Co-reference resolution

  "The man went near the dog. It bit him."

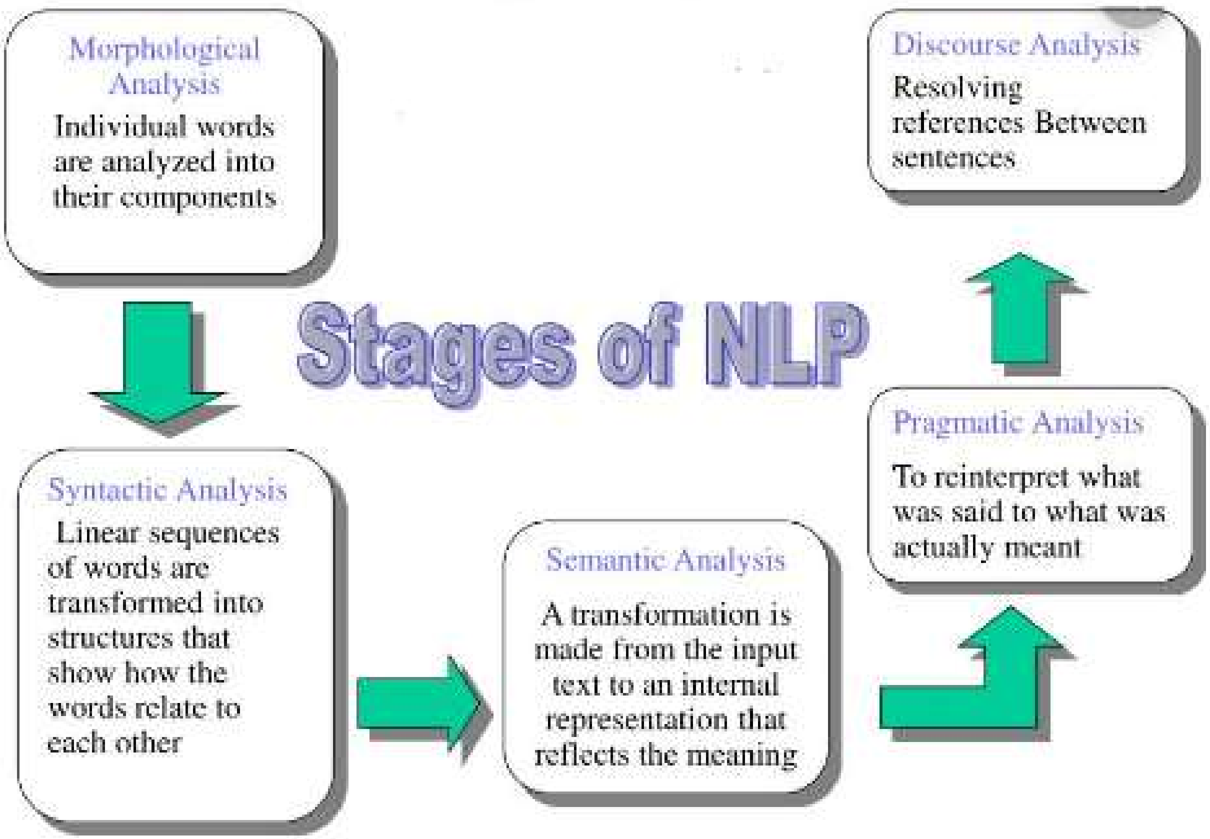
Often co reference & ambiguity go together as in –

"The dog went near the cat. It bit it."

Steps / Stages in NLP



Stages of NLP



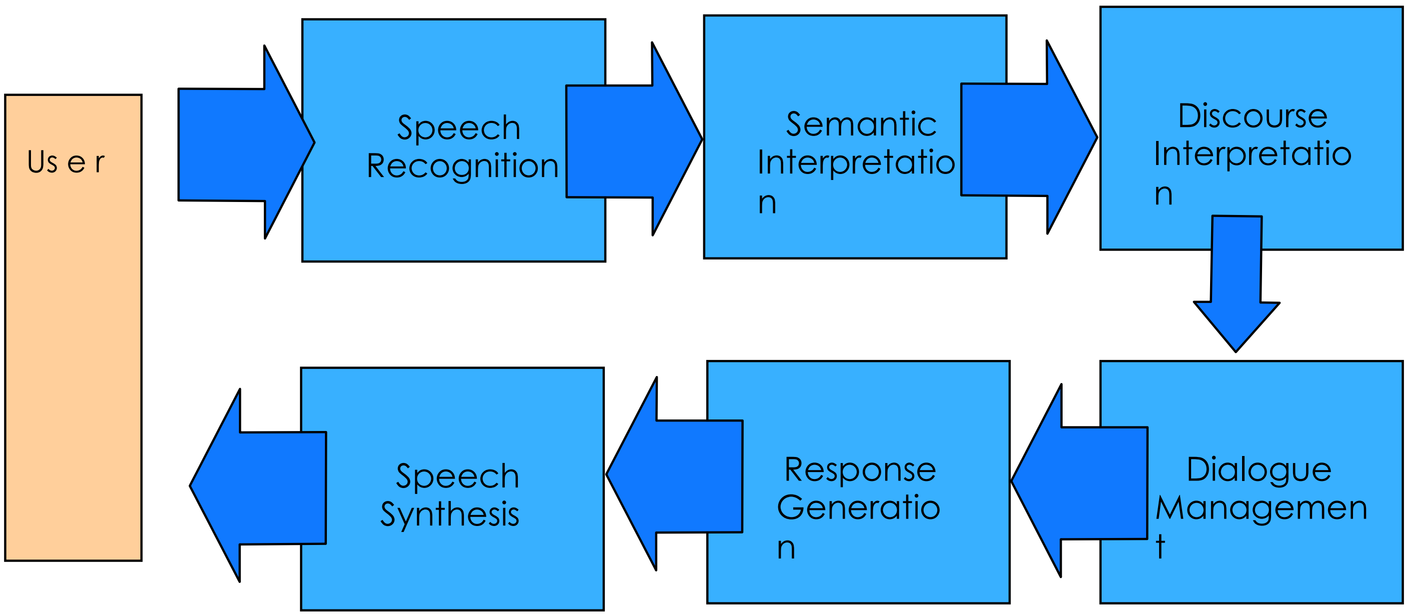
Stages in NLP

- Morphology: Concerns the way words are built up from smaller meaning bearing units. (come(s), co(mes))
- Syntax: concerns how words are put together to form correct sentences and what structural role each word has.
- Semantics: concerns what words mean and how these meanings combine in sentences to form sentence meanings.
- Pragmatics: concerns how sentences are used in different situations and how use affects the interpretation of the sentence.
- Discourse: concerns how the immediately preceding sentences affect the interpretation of the next sentence.

Simple Applications

- Word counters (wc in UNIX)
- Spell Checkers, grammar checkers
- Predictive Text on mobile handsets

Spoken Dialogue System



Parts of the Spoken Dialogue System

- Signal Processing: Convert the audio wave into a sequence of feature vectors.
- Speech Recognition: Decode the sequence of feature vectors into a sequence of words.
- Semantic Interpretation: Determine the meaning of the words.
- Discourse Interpretation: Understand what the user intends by interpreting utterances in context.
- Dialogue Management: Determine system goals in response to user utterances based on user intention.
- Speech Synthesis: Generate synthetic speech as a response.

Stages in NLP System

- Tokenization
- Morphological Analysis
- Syntactic Analysis
- Semantic Analysis (lexical and compositional)
- Pragmatics and Discourse Analysis
- Knowledge-Based Reasoning
- Text generation

Stages of NLP : Tokenization (1)

German:

Lebensversicherungsgesellschaftsangesteller

English:

life insurance company employee

NLP Stages : Morphological / Lexical Analysis (2)

- It involves identifying and analyzing the structure of words.
- Lexicon of a language means the collection of words and phrases in a language.
- Lexical analysis is dividing the whole chunk of txt into paragraphs, sentences, and words.

NLP Stages : Morphological / Lexical Analysis (2)

- Hebrew (transliterated):

  ukshepagashtihu

- English:

  and when I met you (masculine)

NLP Stages : Syntactic Analysis (3)

- Syntax concerns the proper ordering of words and its effect on meaning
- Involves analysis of the words in a sentence to depict the grammatical structure of the sentence.
- The words are transformed into structure that shows how the words are related to each other
- Eg.: "the girl the go to the school" – This would definitely be rejected by the English syntactic analyzer

NLP Stages : Syntactic Analysis (3)

How many readings do the following examples have?

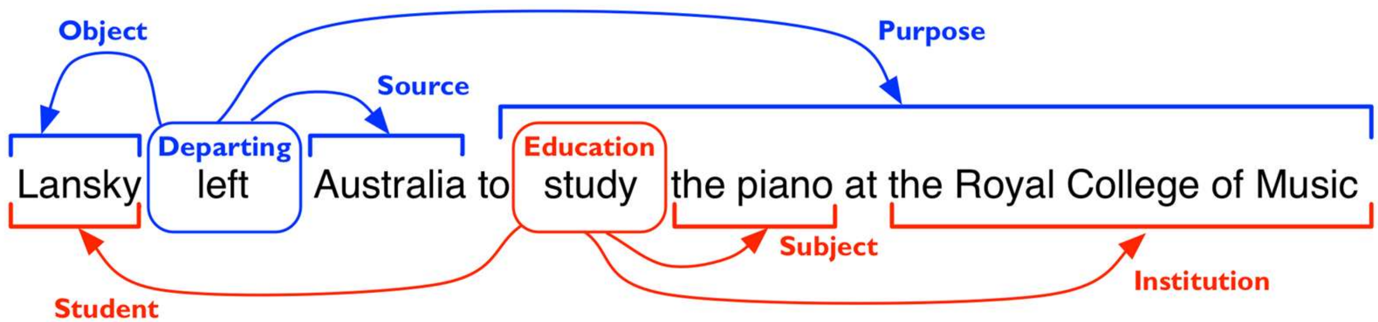
- I made her duck
- I saw Grand Canyon flying to San Diego
- the a are of I
- the cows are grazing in the meadow
- John saw Mary
- Foot Heads Arms Body

NLP Stages : Semantic Analysis (4)

- Semantics concerns the literal meaning of words, phrases and sentences.
- It abstracts the dictionary meaning or the exact meaning from context.
- The structures which are created by the syntactic analyzer are assigned meaning.
- Eg.: "colorless blue idea" – This would be rejected by the analyzer as colorless blue doesn't make any sense together

NLP Stages : Semantic Analysis (4)

- Word sense disambiguation
- Semantic role labeling



Credit: Ivan Titov

NLP Stages : Discourse Analysis (5)

- The structures which are created by the syntactic analyzer are assigned meaning.
- Sense of the context.
- The meaning of any single sentence depends upon the sentences that precedes it and also invokes the meaning of the sentences that follow it.
- Eg.: the word “it” in the sentence, “she wanted it ” – depends upon the prior discourse context.

NLP Stages : Discourse Analysis (5)

- Anaphora resolution -- to resolve referring expression
  - Mary bought a book for Kelly. **She** didn't like **it**.
    - **She** refers to Mary or Kelly. -- possibly Kelly
    - **It** refers to what -- book.
- Discourse structure may depend on application.
  - Monologue
  - Dialogue
  - Human-Computer Interaction

NLP Stages : Discourse Analysis (5)

Q: [Chris] = [Mr. Robin] ?

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh.** As **a boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

Slide modified from Dan Roth

NLP Stages : Discourse Analysis (5)

Co-reference Resolution

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

NLP Stages : Pragmatic Analysis (6)

- Pragmatics concerns the overall communicative and social context and its effect on interpretation.
- It means abstracting or deriving the purposeful use of the language in situations.
- Importantly those aspects of language, which require world knowledge.
- The main focus is on what was said is reinterpreted on what it actually means
- Eg.: “Close the window?” should have been interpreted as a request rather than an order.

Applications of NLP

- Machine Translation
- Information Retrieval

Selecting from a set of documents the ones that are relevant to a query

- Text Categorization

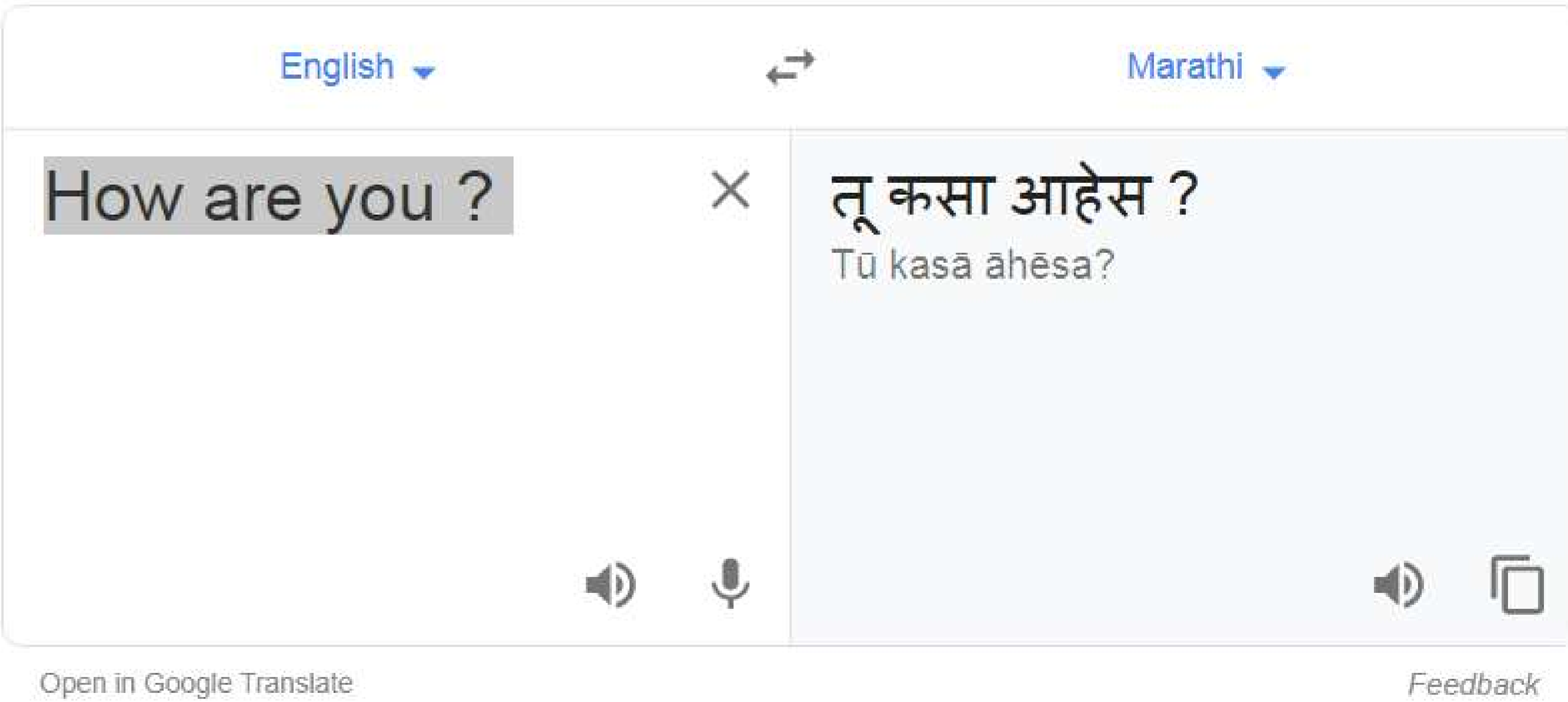
Sorting text into fixed topic categories

- Text summarization
- Extracting data from text : Information Extraction

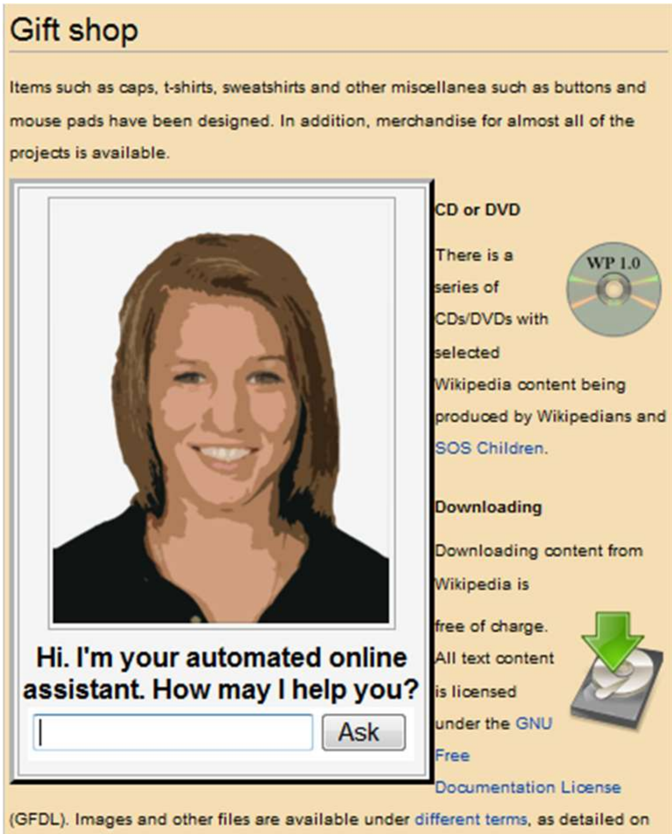
Converting unstructured text into structure data

- Speech Recognition – Google Assistant and Apple Siri

NLP Applications: Machine Translation

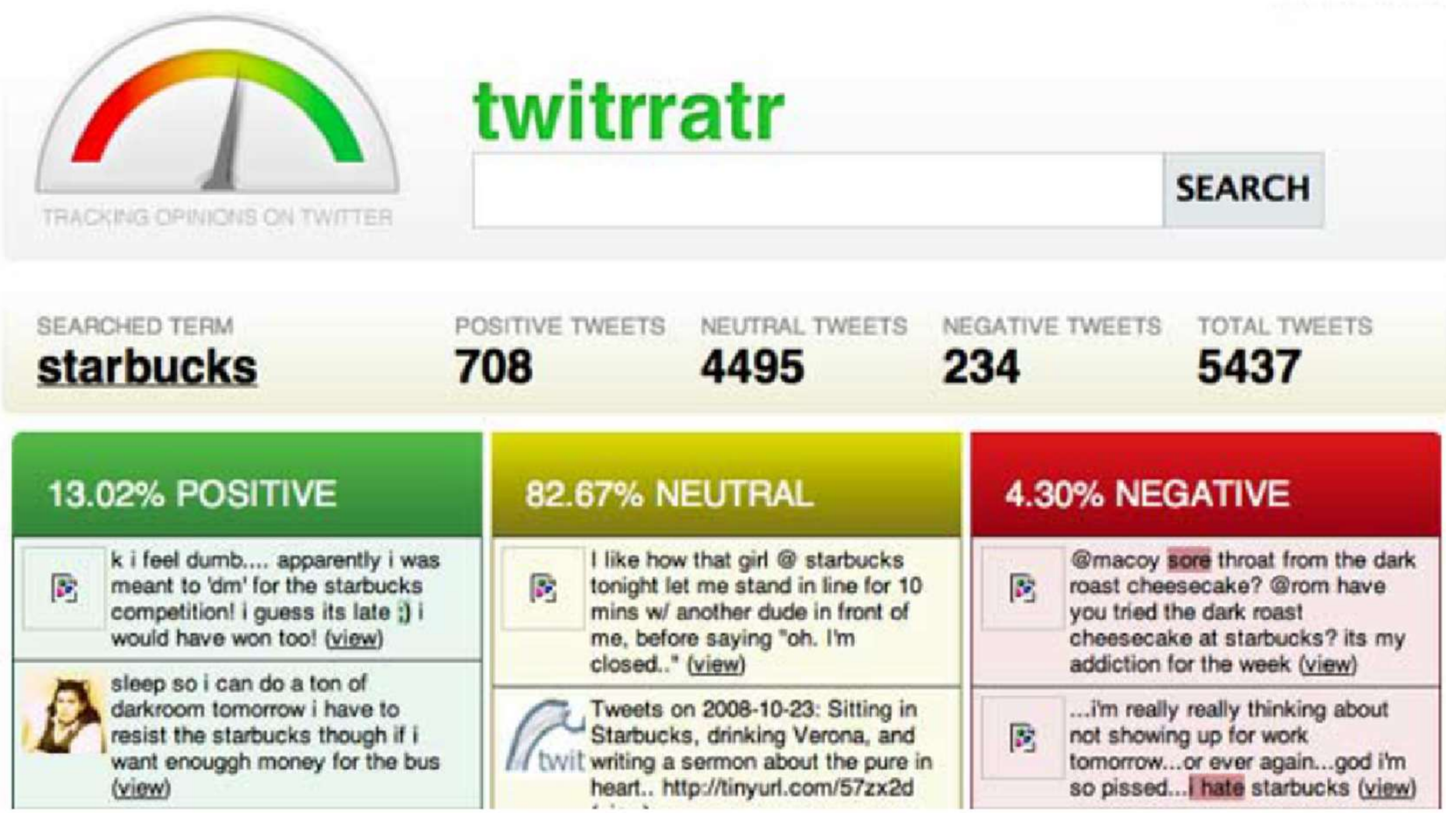


NLP Applications: Dialog Systems

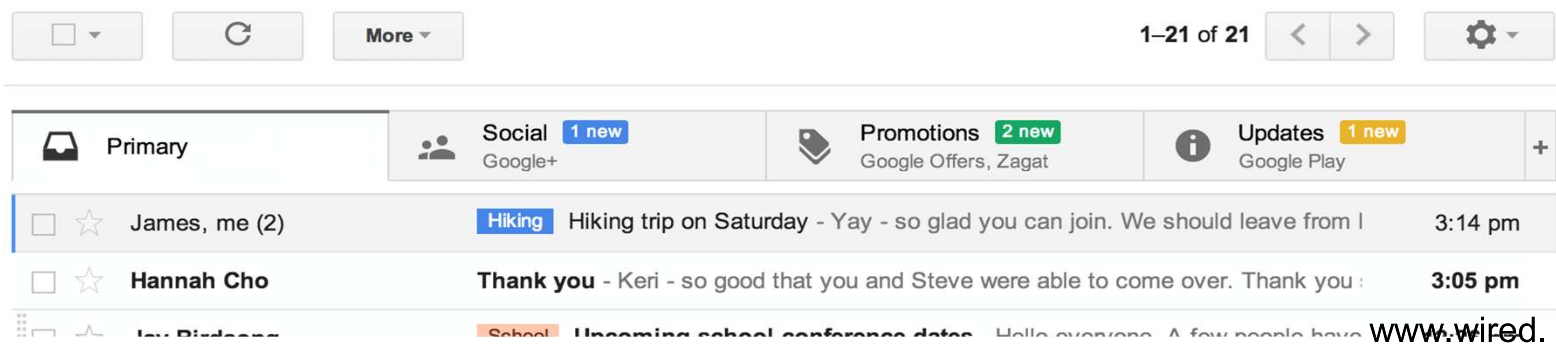


- A **dialogue system**, or conversational agent (CA), is a Computer **system** intended to converse with a human.
- **Dialogue systems** employed one or more of text, speech, graphics, haptics, gestures, and other modes for communication on both the input and output channel.

NLP Applications: Sentiment / Opinion Analysis



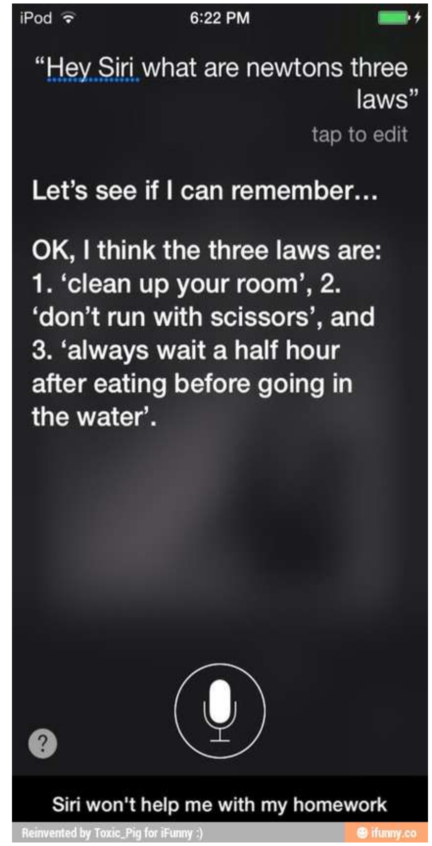
NLP Applications: Text Classification



www.wired.com

NLP Applications: Question Answering

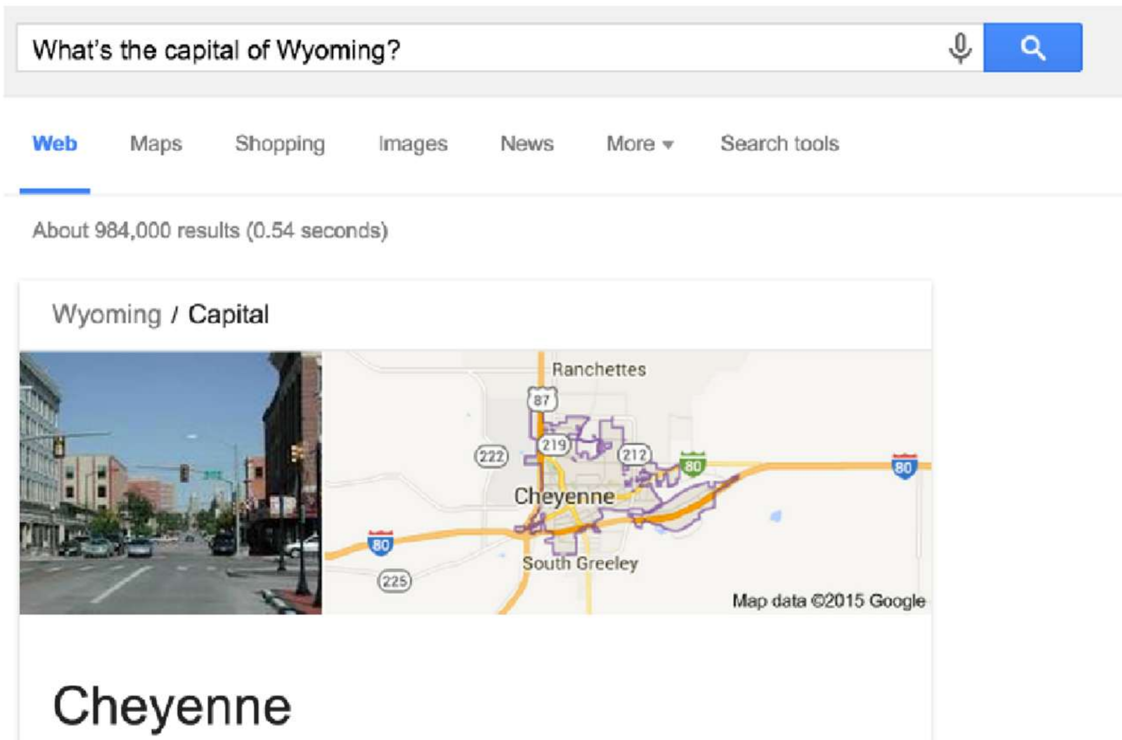
['Watson' computer wins at 'Jeopardy']



credit: ifunny.com

NLP Applications: Question Answering

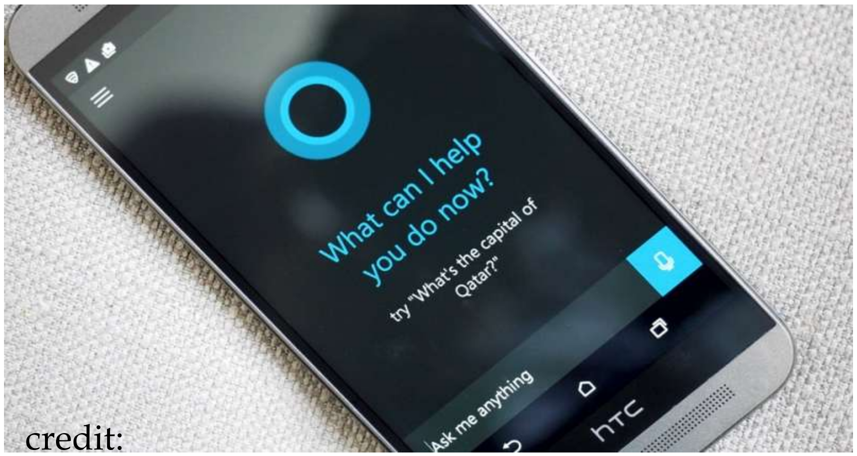
Go beyond search…..



NLP Applications: Natural Language Instruction

NLP Applications: Digital Personal Assistant

More on natural language



credit: techspot.com

- Semantic parsing – understand tasks
- Entity linking – “my wife” = “Kellie” in the phone book

NLP Applications: Information Extraction

Unstructured text to database Entries

New York Times Co. named Russell T. Lewis, 45, president and general manager of its flagship New York Times newspaper, responsible for all business-side activities. He was executive vice president and deputy general manager. He succeeds Lance R. Primis, who in September was named president and chief operating officer of the parent.

Person	Company	Post	State
Russell T. Lewis	New York Times newspaper	president and general manager	start
Russell T. Lewis	New York Times newspaper	executive vice president	end
Lance R. Primis	New York Times Co.	president and CEO	start

NLP Applications: Language Comprehension

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh.** As **a boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

- **Q: who wrote Winnie the Pooh?**
- **Q: where is Chris lived?**

Some NLP Applications

- Finding appropriate documents on certain topics from a database of texts (for example, finding relevant books in a library)
- Extracting information from messages or articles on certain topics (for example, building a database of all stock transactions described in the news on a given day)
- Translating documents from one language to another (for example, producing automobile repair manuals in many different languages)
- Summarizing texts for certain purposes (for example, producing a 3-page summary of a 1000-page government report)

Some NLP Applications

- question-answering systems, where natural language is used to query a database (for example, a query system to a personnel database)

- automated customer service over the telephone (for example, to perform banking transactions or order items from a catalogue)

- tutoring systems, where the machine interacts with a student (for example, an automated mathematics tutoring system)

- spoken language control of a machine (for example, voice control of a VCR or computer)

Why NL Understanding is Hard?

- Natural language is extremely rich in form and structure, and **very ambiguous**.
  - How to represent meaning,
  - Which structures map to which meaning structures.
- One input can mean many different things. Ambiguity can be at different levels.
  - Lexical (word level) ambiguity -- different meanings of words
  - Syntactic ambiguity -- different ways to parse the sentence
  - Interpreting partial information -- how to interpret pronouns
  - Contextual information -- context of the sentence may affect the meaning of that sentence.
- Many input can mean the same thing.
- Interaction among components of the input is not clear.

Challenges of NLP : Ambiguity

- Ambiguity occurs when a word, phrase, or sentence has multiple interpretations.
- Humans resolve ambiguity using context and knowledge.
- Machines find ambiguity challenging.
- Ambiguity resolution at all levels and in all systems components is one of the major tasks for NLP.

Importance of Ambiguity in NLP

- Affects machine translation
- Impacts chatbots & QA systems
- Impacts information retrieval
- Complicates semantic understanding
- More complex in multilingual NLP and morphologically rich languages

Lexical Ambiguity – WSD : NLP Challenge

Word with multiple meanings

- **English:**
- bank – river/finance (**Eg.:** He sat near the bank.→ Meaning depends on context.)
- bat – animal/equipment
- Light – illumination / not heavy
- **Indian Regional Languages:**
- Hindi: कल (yesterday/tomorrow), हार (haar) → defeat / necklace
- Marathi: आम (mango/common)
- Tamil: கால் (kaal) → leg / quarter / time

Syntactic Ambiguity (Structural Ambiguity)

- Sentence structure allows multiple interpretations (grammatical parses).
- English:
- I saw the man with a telescope.
- Possible meanings: I used a telescope to see the man /
- The man had a telescope

Syntactic Ambiguity (Structural Ambiguity)

- Sentence structure allows multiple interpretations (grammatical parses).
- Hindi: (Maine durbeen se aadmi ko dekha)
- मैंने दूरबीन से आदमी को देखा।
- Ambiguity: Instrument attached to verb / Instrument attached to noun
- Indian languages often allow flexible word order, increasing syntactic ambiguity.
- **NLP Challenge:** Parsing and tree selection

Semantic Ambiguity

- Occurs when sentence meaning is unclear despite correct grammar.
- **English:**
- Visiting relatives can be boring.

Interpretations:

- Relatives who visit are boring.
- The act of visiting relatives is boring

Semantic Ambiguity

- **Tamil:**
- படிக்கும் மாணவர்கள் புத்திசாலிகள்

Interpretations:

- Students who study are intelligent
- Studying students are intelligent (temporary state)
- **NLP Challenge:** Representing meaning using logic or embeddings

Pragmatic Ambiguity

- Occurs due to context, intention, or real-world knowledge.

English:

- Can you pass the salt?
- Literal meaning: Ability question
- Intended meaning: Request

Pragmatic Ambiguity

- **Hindi:** (kya aap darwaaza band kar sakte hein?)
- क्या आप दरवाज़ा बंद कर सकते हैं?
- Meaning:
- Ability inquiry ❌
- Polite command ✅
- **NLP Challenge:** Intent detection and dialogue understanding

Morphological Ambiguity

- Single word expresses multiple grammatical meanings.
- Occurs in languages with rich inflectional morphology.
- Marathi:
- लिखतो (likhto) – I write / He writes (context-dependent)
- Tamil: வந்தான் (vandhaan)
- Verb tense + gender + number encoded in one word
- NLP Challenge: Morphological analysis in low-resource languages

Discourse Ambiguity

- Occurs when meaning depends on previous sentences.
- English:
- John told Ravi that he was late.
- Who was late? John / Ravi
- Hindi: राम ने श्याम से कहा कि वह देर से आया।
- (ram ne shyaam se kahan ki woh der se aaya)
- Pronoun ambiguity common due to pronoun dropping.
- NLP Challenge: Coreference resolution

Ambiguity Comparison: English Vs Regional Languages

Aspect	English	Indian Languages
Word Order	Fixed (SVO)	Free / flexible
Morphology	Relatively simple	Rich & complex
Script	Single (Latin)	Multiple scripts
Pronoun Usage	Explicit	Often implicit
Ambiguity Level	Moderate	High

Challenges of NLP : Translation

"The coach lost a set"

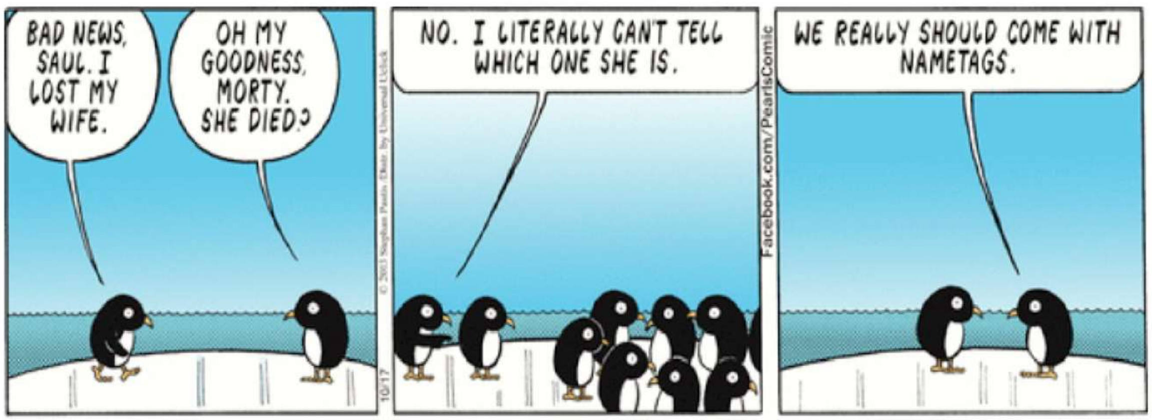
One strongly preferred meaning although in a standard English-Russian dictionary

- coach has 15 senses
- lose has 11 senses
- set has 91 sense

15 x 11 x 91 = 15015 possible translations

Challenges of NLP : Ambiguity

- Word sense ambiguity



credit: A. Zwicky

Challenges of NLP : PP Ambiguity


San Jose cops kill man with knife

Text Paper Translate Listen Close

San Jose cops kill man with knife

Ex-college football player, 23, shot 9 times allegedly charged police at fiancee's home

By Hamed Aleaziz and Vivian Ho

A man fatally shot by San Jose police officers while allegedly charging at them with a knife was a 23-year-old former football player at De Anza College in Cupertino who was distraught and depressed, his family said Thursday.

Police officials said two officers opened fire Wednesday afternoon on Phillip Watkins outside his fiancee's home because they feared for their lives. The officers had been drawn to the home, officials said, by a 911 call reporting an armed home invasion that, it turned out, had been made by Watkins himself.

But the mother of Watkins' fiancee, who also lives in the home on the 1300 block of Sherman Street, said she witnessed the shooting and described it as excessive. Faye Buchanan said the confrontation happened shortly after she called a suicide intervention hotline in hopes of getting Watkins medical help.

Watkins' 911 call came in at 5:01 p.m., said Sgt. Heather Randol, a San Jose police spokeswoman. "The caller stated there was a male breaking into his home armed with a knife," Randol said. "The caller also stated he was locked in an upstairs bedroom with his children and requested help from police."

She said Watkins was on the sidewalk in front of the home when two officers got there. He was holding a knife with a 4-inch blade and ran toward the officers in a threatening manner, Randol said.

"Both officers ordered the suspect to stop and drop the knife," Randol said. "The suspect continued to charge the officers with the knife in his hand. Both officers, fearing for their safety and defense of their life, fired at the suspect."

On the police radio, one officer said, "We have a male with a knife. He's walking toward us."

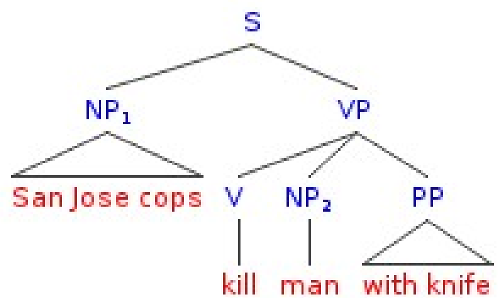
"Shots fired! Shots fired!" an officer said moments later.

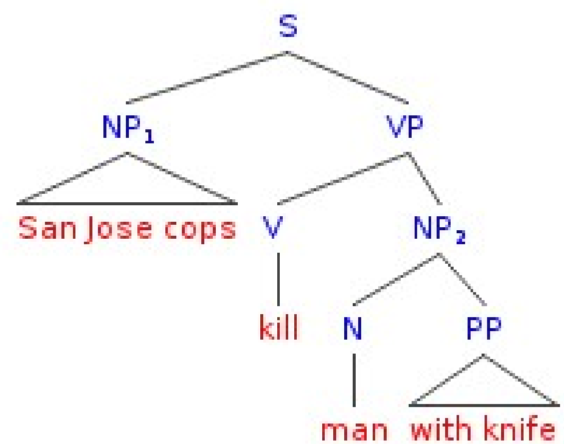
A short time later, an officer reported, "Male is down. Knife's still in hand."

Buchanan said she had been prompted to call the

Shoot continues on D8

Back Continue





Credit: Mark Liberman, http://languagelog.ldc.upenn.edu/nll/?p=17711

Challenges of NLP : Ambiguity Headlines

- Include your children when baking cookies
- Local High School Dropouts Cut in Half
- Hospitals are Sued by 7 Foot Doctors
- Iraqi Head Seeks Arms

- Safety Experts Say School Bus Passengers Should Be Belted

- Teacher Strikes Idle Kids

Challenges of NLP : Pronoun Reference Ambiguity

Credit: http://www.printwand.com/blog/8-catastrophic-examples-of-word-choice-mistakes

Challenges of NLP : Language is not Static

- Language grows and changes

LOL	Laugh out loud
G2G	Got to go
BFN	Bye for now
B4N	Bye for now
Idk	I don’t know
FWIW	For what it’s worth
hru	How are you

Resolve Ambiguities

Models and *algorithms* to resolve ambiguities at different levels:

- **Part-of-Speech tagging** -- Deciding whether duck is verb or noun.
- **Word-Sense disambiguation** -- Deciding whether make is create or cook.
- **Lexical disambiguation** -- Resolution of part-of-speech and word-sense ambiguities are two important kinds of lexical disambiguation.
- **Syntactic ambiguity** -- her duck is an example of syntactic ambiguity, and can be addressed by probabilistic parsing.

Handling Ambiguity in NLP Systems

- Rule-based methods
- Statistical models
- Machine learning
- Contextual embeddings (BERT, mBERT)
- Knowledge graphs
- Transformer-based disambiguation